

How to Run the Project

Follow these steps to set up and run the ai_demand_supply_balancer project on your local machine.

Prerequisites

- **Node.js:** Ensure you have Node.js installed (version 18 or higher is recommended).
- **npm:** Comes installed with Node.js.

1. Unzip the Project

Extract the contents of ai_demand_supply_balancer.zip to a folder on your computer.

2. Install Dependencies

Open your terminal (Command Prompt, PowerShell, or Terminal), navigate to the project folder, and run:

```
npm install
```

This will install all the necessary libraries listed in package.json.

3. Environment Setup

The project requires environment variables to connect to the database and handle authentication.

A .env file is included in the zip with default/development values.

Ensure the .env file exists in the root directory with the following content structure:

```
MONGODB_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
NEXTAUTH_URL=http://localhost:3000
NEXTAUTH_SECRET=your_nextauth_secret
```

Note: The included .env file should work out-of-the-box for this project if the database credentials are still valid.

4. Database Seeding (Optional but Recommended)

To populate the database with initial dummy data, run:

```
npm run seed
```

5. Run the Development Server

To start the application in development mode:

```
npm run dev
```

Open <http://localhost:3000> (<http://localhost:3000>) with your browser to see the result.

6. Build for Production (Optional)

To build the application for production usage:

```
npm run build  
npm start
```